

Anchoring Critic-Guided Fine-Tuning of Generative Robot Policies

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Abstract—Reinforcement learning can improve robot policies learned from demonstrations, but errors in a learned critic can also destroy their existing skills. We present CAST (Constrained Action-Space Transport), a method for fine-tuning generative robot policies using task-success rewards. CAST learns corrections to a frozen pretrained policy: it adjusts sampled actions using the critic and trains a small network to reproduce them. A restoring term permits small corrections while resisting larger departures from the pretrained behavior. On eight challenging LIBERO-90 tasks, per-task fine-tuning raises mean success from 66.1% to 75.5%, compared with 67.0% for the standard residual actor update. On robomimic square, success rises from 38.2% to 91.2–93.0%. Removing the restoring term causes zero success on all five LIBERO ablation tasks, even with step clipping. The key is to limit accumulated departure from the pretrained policy, not merely the size of each update.

I. Introduction

Learning from demonstrations gives robots a useful starting point, but a policy that usually succeeds may still need practice to become reliable. Reinforcement learning can provide that practice without requiring additional demonstrations, and recent methods apply it to generative policies that produce sequences of robot actions [1], [2]. The challenge is to improve an existing skill while preserving the behavior that already works. We study this problem when feedback is limited to task success, so failed attempts provide little information about how to recover.

A practical approach is residual fine-tuning: freeze the pretrained policy and learn a correction to its actions, guided by a learned critic [3]. The critic predicts the return of an action, and the correction network is trained to increase that prediction. However, critic accuracy is limited by the data available during training. Optimizing its predictions can exploit errors outside well-covered regions of the action space [4], [5]. The policy may then move away from successful pretrained behavior even as its predicted return increases. With success-only rewards, losing that behavior also reduces the successful experience available to correct the critic.

This raises a concrete design question: should fine-tuning limit the size of each update, or the accumulated departure from the pretrained policy? These are different constraints. Many small updates can move a policy far from its starting point. A restoring term tied to a frozen reference instead opposes that accumulated departure. Yet restoring too strongly everywhere can suppress the corrections needed to improve. The desired behavior is to permit local changes while resisting larger ones. Fig. 1 illustrates why this distinction matters.

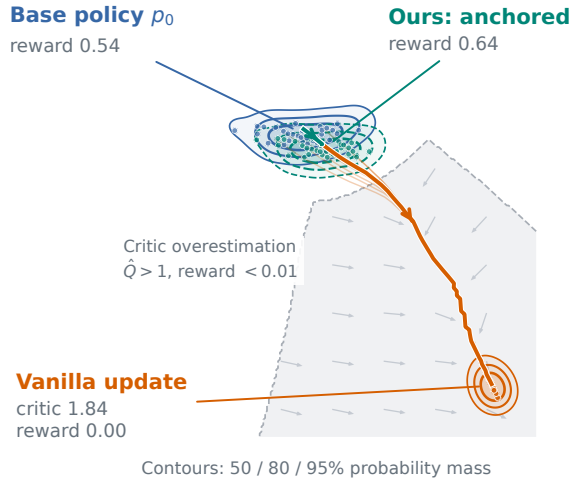


Fig. 1. Distribution drift under a misleading critic. Blue shows an analytic base policy; orange and green show 384 paired particles after 60 critic-ascent updates without and with anchoring. Contours enclose 50/80/95% of the estimated probability mass; dots and paths show representative particles. Gray arrows follow increasing critic value in an overestimated region. Scores are particle averages. This single-seed toy uses 25 critic-training samples and a 0.06 step cap. Projection within 0.15 of each particle’s base action illustrates CAST’s anchoring principle, rather than full policy training.

We present CAST (Constrained Action-Space Transport), a residual fine-tuning method that implements this principle. At each update, it samples actions from the frozen base plus the current correction, moves them along the critic’s action gradient, and trains the correction network to reproduce the adjusted actions. This action-improvement approach builds on prior critic-guided training of generative policies [6], [7]. CAST combines displacement clipping with a restoring field toward the corresponding frozen-base action. A dead zone in this field leaves small corrections unpenalized by that term and pulls back larger departures. The adjusted actions become fixed regression targets, so the actor update requires neither policy likelihoods nor differentiation through the frozen generator.

We evaluate the method on simulated manipulation tasks from LIBERO and robomimic, using matched base policies, critic-training procedures, and interaction budgets for the residual-actor comparison. On eight challenging LIBERO-90 tasks, per-task fine-tuning improves mean success by 9.4 percentage points, compared with 0.9 points for the standard DICE-RL actor. On robomimic square, CAST reaches 91.2–93.0% success

from a 38.2% base, compared with 86.1–90.9% for the standard actor at the reported budget. These gains are conditional: staying near the base can also limit improvement when a task requires substantially different actions.

Our five-task LIBERO ablation isolates the two constraints. Removing all base-restoring terms causes zero success, even with step clipping. Restoration alone avoids complete collapse but is inconsistent across tasks; the full combination improves on every task’s base. A dead-zone penalty in the standard actor’s loss also enables improvement. These results identify restoring geometry as a useful principle shared by both actor updates.

Our contributions are:

- A controlled study distinguishing per-update clipping from fixed-base restoration in critic-guided robot policy fine-tuning.
- CAST, a likelihood-free residual update that combines critic-guided action targets with clipping and base-restoring fields.
- Manipulation experiments identifying the benefits and limits of anchoring, including its transfer to a standard actor’s loss.

II. Related Work

A. Fine-Tuning Generative Robot Policies

Generative policies require specialized reinforcement learning updates because their action likelihoods are often intractable. DPPO optimizes a diffusion policy through its denoising process [1], while ReinFlow introduces stochastic transitions into flow sampling to enable policy-gradient updates [2]. Flow Matching Policy Gradients uses differences in flow-matching losses to construct a policy-update surrogate [8]. These methods exploit the structure of the generator or its training objective. In contrast, CAST needs only samples from a frozen noise-to-action map and a critic. This makes its residual update applicable to one-step generators as well as iterative samplers. Our focus is how to preserve pretrained behavior during critic-guided improvement.

B. Critic-Guided Action Improvement

Using a critic to improve actions before fitting a policy is an established approach. DIPO modifies replay-buffer actions by gradient ascent and trains a diffusion policy on the resulting targets [6]; Q-Score Matching trains a diffusion policy using the critic’s action gradient [7]. Particle-based methods provide another connection: Soft Q-Learning trains an amortized sampler using Stein variational updates [9], and drifting models train one-step generators by fitting displaced samples [10]. Drifting Field Policy uses critic-ranked candidates to guide updates on a drifting backbone [11]. CAST builds on these ideas by constructing fresh action targets for a residual on a frozen base, with displacement clipping and explicit restoration toward the base. Stored transitions are left unchanged.

An alternative is to use critic values without following their action gradients. AWR and MPO favor actions through advantage- or value-weighted fitting [12], [13], while IDQL combines a diffusion behavior model with critic-based action selection [14]. These approaches motivate our ranking-only and value-weighted field variants. We compare them with the gradient field to distinguish the value of the improvement signal from the effect of anchoring.

C. Residual Learning and Pretrained-Policy Anchoring

Residual learning preserves a base controller while learning corrections. DICE-RL applies this construction to generative policies, combining a critic-driven residual actor with behavior-cloning regularization [3]. It is our closest system-level reference and the source of our controlled actor comparison. DAWN studies the critic cold-start and scaling issues that affect residual RL [15]. More broadly, TD3 and BCQ show how function-approximation and extrapolation errors can undermine critic-guided optimization [4], [5].

Our contribution is a controlled study of the restoring constraint: limiting each displacement, pulling toward a fixed base, and allowing a region of small corrections have different effects. The successful loss-space hinge control is important to this positioning. It shows that useful restoring geometry can be expressed within a conventional actor loss as well as through CAST’s action targets. We therefore study when that geometry preserves and improves pretrained skills, rather than claiming that target regression alone resolves critic error.

III. Problem Statement

A. Setting

We consider online fine-tuning of a generative robot policy pretrained from demonstrations. An observation s includes the inputs used by the policy, and an action $a \in \mathbb{R}^{Hd}$ is a chunk of H controls with d dimensions each. A frozen generator π_0 maps s and noise z to an action. Following residual RL [3], we learn a correction:

$$a_\theta(s, z) = \pi_0(s, z) + r_\theta(s, z), \quad z \sim \mathcal{N}(0, I). \quad (1)$$

An ensemble critic estimates return from replayed experience. Reward is provided on task success. The goal is to increase success through online interaction while retaining useful pretrained behavior. Our controlled comparison varies the actor update while matching the base, critic-training procedure, and interaction budget.

B. The Problem: Accumulated Departure

A critic can assign high values to actions outside its training coverage. Repeatedly following those predictions may move the policy away from actions that actually succeed [5]. A step cap alone does not prevent this: $\|a_{t+1} - a_t\| \leq \delta$ permits distance from the initial action to grow with the number of updates. Here t indexes optimization steps, not control steps. Fig. 1 follows an entire

base action distribution through paired particle updates. Unanchored critic ascent moves its probability mass into an overestimated region; anchoring each particle to its own base action permits improvement nearby.

The design question is how to permit useful corrections without allowing uncontrolled departure from the frozen base. We measure this departure by the residual’s root-mean-square action magnitude, $D_{\text{base}} = \mathbb{E}_{s,z}[\|r_\theta(s, z)\|_2/\sqrt{Hd}]$, with states sampled from replay. A fixed-base restoring term acts on this departure, whereas step clipping limits only each proposed displacement. We study both empirically: bounded regression targets do not themselves guarantee bounded outputs after fitting a shared neural network.

IV. Method

CAST improves sampled actions and trains the residual to reproduce them. Each target combines a critic-guided step with a restoring correction toward the frozen base. This gives the update two separate controls: clipping limits the proposed step, while restoration opposes accumulated departure. All operations below use normalized action coordinates, with chunk dimension $D = Hd$.

A. Critic-Guided Proposals

For each replay state s , draw K noise samples and form current actions:

$$\bar{r}_i = \text{sg}[r_\theta(s, z_i)], \quad a_i = \pi_0(s, z_i) + \bar{r}_i, \quad (2)$$

where sg denotes stop-gradient. Let $\bar{Q}_\phi(s, z, a)$ be the conservative critic-ensemble aggregate; the noise input is omitted when the critic does not use it. Following the baseline’s value normalization [3], divide by a detached scale $q_{\text{scale}} = \max(\epsilon, \mathbb{E}_{\text{batch}}[\|\bar{Q}_\phi\|])$, computed over the current candidates. The reward displacement is

$$d_i^Q = \eta_Q C_{\delta_Q} \left(\frac{\nabla_a \bar{Q}_\phi(s, z_i, a_i)}{q_{\text{scale}}} \right), \quad (3)$$

where $C_\delta(v) = v \min(1, \delta/\|v\|_2)$, with $C_\delta(0) = 0$. The clip rescales large gradients while preserving small ones; η_Q then sets the reward step size. We differentiate the critic with respect to its action input only, holding its parameters fixed during target construction.

B. Restoration Toward the Frozen Base

Each candidate is anchored to the base action generated by the same (s, z_i) . Define its residual magnitude in root-mean-square units as $m_i = \|\bar{r}_i\|_2/\sqrt{D}$. The restoring displacement is

$$d_i^R = -\eta_0 \bar{r}_i - \eta_{\text{anc}} \left(1 - \frac{\rho}{\max(m_i, \epsilon)} \right)_+ \bar{r}_i, \quad (4)$$

where $(x)_+ = \max(x, 0)$. The first term supplies a weak, always-on pull. The second is inactive within radius ρ and opposes the excess displacement outside it. Measuring ρ in rms units makes it comparable across

action-chunk dimensions. The reference remains frozen throughout training, so the restoring force responds to total departure rather than just the latest step.

This is a soft correction evaluated at the current residual, rather than the hard projection used for illustration in Fig. 1. Setting $\rho = 0$ makes both terms always active. Removing restoration requires setting both η_0 and η_{anc} to zero.

C. Regression onto the Adjusted Targets

Combine the two displacements, clip their sum, and detach the target:

$$\tilde{r}_i = \text{sg} \left[\bar{r}_i + C_{\delta_{\text{tot}}}(d_i^Q + d_i^R) \right]. \quad (5)$$

The residual network is then trained by mean-squared error,

$$\mathcal{L}_{\text{actor}} = \mathbb{E}_s \left[\frac{1}{KD} \sum_{i=1}^K \|r_\theta(s, z_i) - \tilde{r}_i\|_2^2 \right]. \quad (6)$$

Only this regression loss updates the actor. The base stays frozen, and fresh targets are constructed without overwriting stored transitions. The target displacement satisfies $\|\tilde{r}_i - \bar{r}_i\|_2 \leq \delta_{\text{tot}}$. As noted in the problem statement, fitting these targets with a shared network does not guarantee the same bound on realized policy changes.

The dead-zone field is proportional to the negative gradient of a hinged radial penalty, $(\|r\|_2 - \sqrt{D}\rho)_+^2$. Its geometry can therefore also be placed in a conventional actor loss. Our hinge-loss control tests this connection directly; target regression is an implementation of the restoring principle, not its only possible form.

D. Training and Variants

We alternate online collection with critic and actor updates. Critic training retains the baseline’s ensemble temporal-difference updates, n -step returns, and demonstration replay. These settings are matched within each controlled actor comparison. The policy requires no action likelihood, and inference uses the frozen generator plus the residual in (1).

The LIBERO recipe uses $K = 16$, $\eta_0 = 0.05$, $\eta_{\text{anc}} = 1$, $\rho = 0.05$ rms, $\delta_Q = 1$, and $\delta_{\text{tot}} = 0.15$. We compare shared reward steps $\eta_Q \in \{1, 2\}$ across tasks. The robomimic recipe uses $\eta_Q = 0.5$ and $\rho = 0$. For reward-field ablations, we replace the action gradient with particle transport [10] toward either the critic’s top- k candidates or candidates weighted by a softmax of standardized Q -values. The clipping, restoration, and regression steps remain the same, isolating how much the update benefits from the critic’s gradient direction.

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